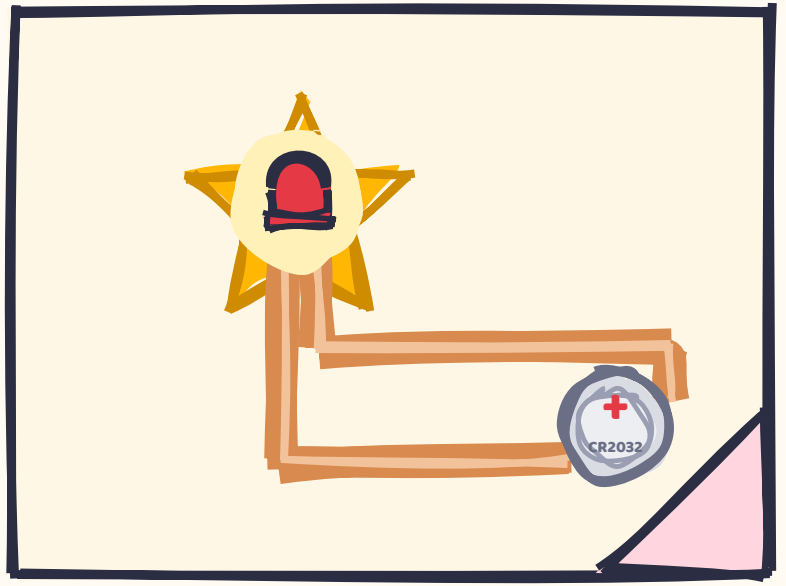
Little Explorers · Class 1

Light-Up Paper Circuit Card

Can a piece of paper make light?

Closed circuits | LED + coin cell | Plus and minus

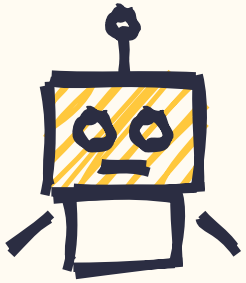


Welcome, Explorers!
Today you build your very first circuit, and it goes home as a gift!

Hello, Explorers!

Welcome to the Robotics & AI Club

In our club we will make **lights**, **sounds**, **buttons** and real **moving robots**. And it all starts today, with one tiny light!



Look and listen
when the teacher shows a new part

Share and help
we build as a team

Gentle hands
parts are tiny and precious

Try again!
mistakes help us learn

Our club promise
We look, we share, we are gentle and we never give up!

LEARN

Magic or science?

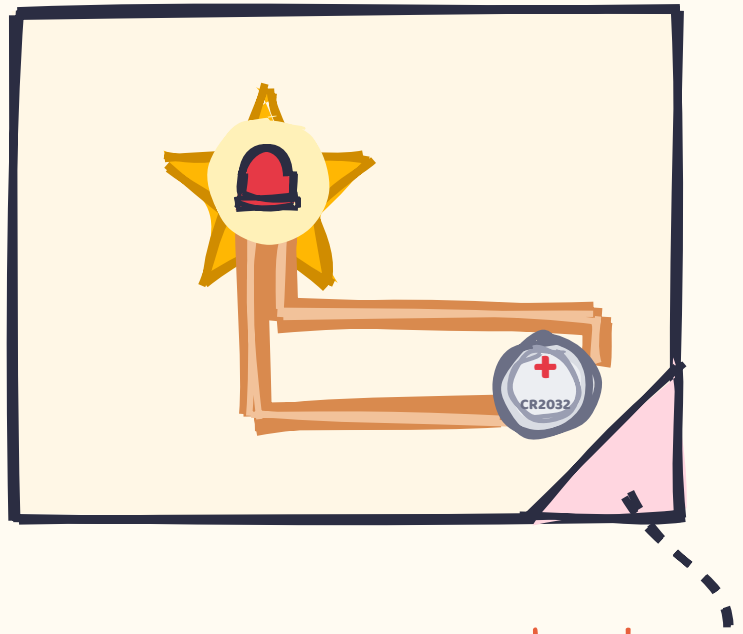
Can paper light up?

Watch your teacher's card. It looks like normal paper… Press the corner and… **it glows!** How?

A magic spell?

A hidden torch?

A secret battery?

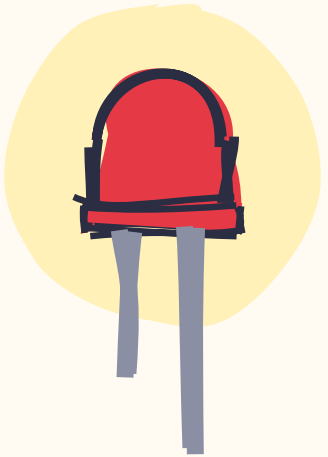


Think, then whisper

Tell your partner your best guess. We find the answer by the end of class!

Our learning aim

Today, I will learn to...



1 I can tell that electricity needs a **closed loop** to flow.

2 I can find the **plus (+)** and **minus (-)** side of an LED and a battery.

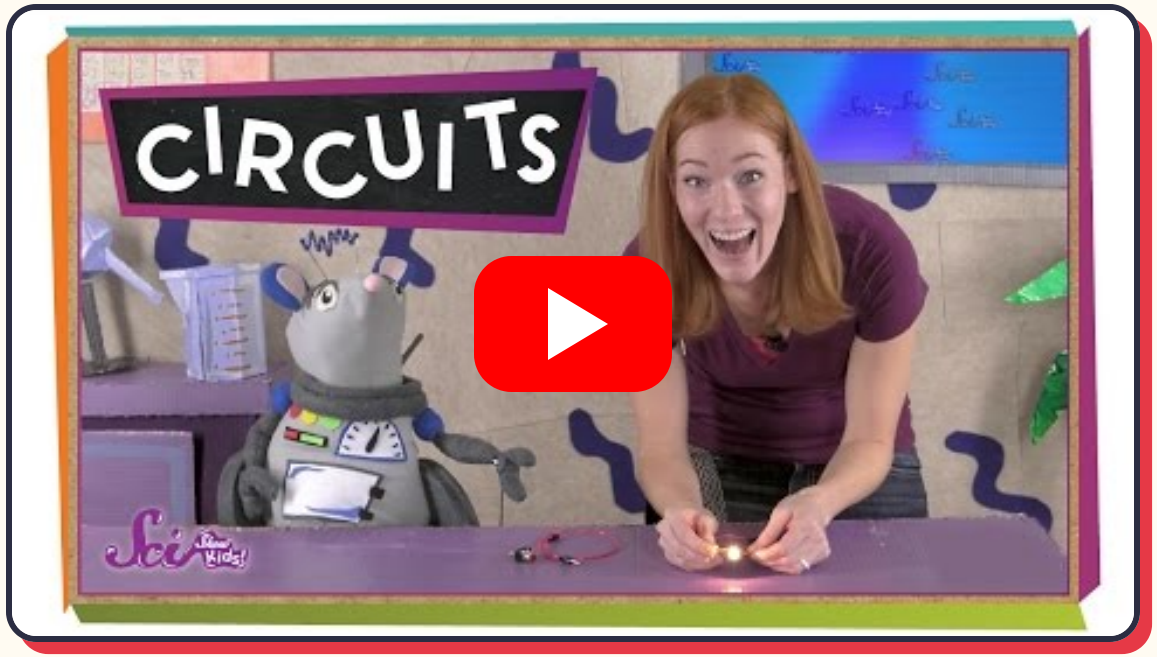
3 I can build a **card that lights up** and explain how it works.

Keywords

circuit · loop · LED · battery · plus · minus · conductor

Watch and think

Video time!



The Power of Circuits!

SciShow Kids · about 5 minutes · youtu.be/HOFp8bHTN30

Watch for...

1. What is a **circuit**?
2. What happens if the circle is **broken**?
3. What gives the circuit its power?

Scan to watch

After the video: Trace a circle in the air with your finger. Where does the electricity go?

Look around you

Electricity is everywhere!



Lights
at home and on the street



Phones
a battery inside



Toy cars
zoom with batteries



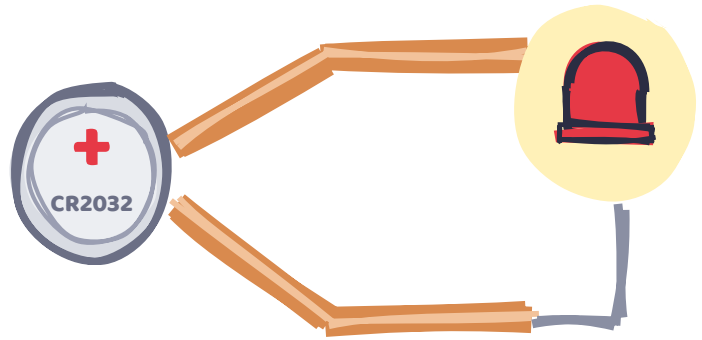
Speakers
make sounds

Electricity is invisible energy that travels along a path. When it reaches a light, the light switches on!

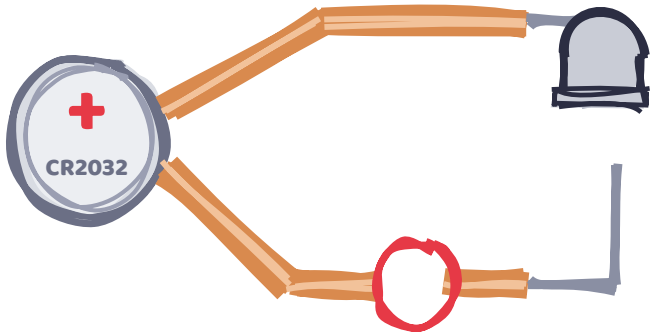
Spot it!
Count how many things in our classroom need electricity!

The big secret

A circuit is a circle road



CLOSED loop: light ON



OPEN loop (gap): light OFF

Electricity leaves the battery, runs along the road, lights the LED and goes **back home** to the battery. No gaps allowed!

LEARN

Every part has two sides

Plus (+) and minus (-)



Long leg = PLUS

Short leg = minus



Writing side = PLUS

Plain side = minus

Say it with me!
Long leg, plus leg! Long leg, plus leg!

Match it
Plus meets plus. Minus meets minus.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Explorer safety rules
2. Be a human circuit!
3. Meet the parts
4. Guess, then test
5. Make your card in 6 steps
6. How your card is wired
7. Be a circuit detective

Before we start

Explorer safety rules

Our tiny coin battery is **safe to touch**. It cannot hurt your fingers.

Never put a battery in your mouth, nose or ears. Ask the teacher to hold it if you need a break.

Never play with wall sockets. Big electricity is only for grown-ups.

Copper tape edges are thin. **Press it gently** with your finger.



coin cell

Teacher check
Count the batteries out, count them back in at the end!

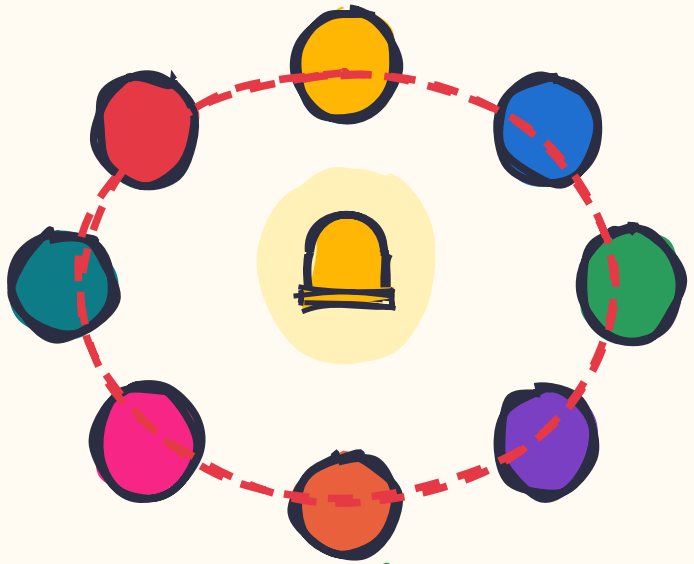
MAKE

Stand up and play!

Be a human circuit!

How to play

1. Stand in a circle and **hold hands**.
2. The teacher is the **battery**. She sends a squeeze.
3. Pass the squeeze all the way round.
4. When it comes back, the child in the middle shines the **torch**!



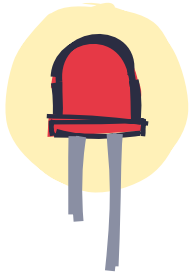
Hold hands: the light stays ON!

? **Now let go!** One friend drops hands. Can the squeeze get through? **No!** The light goes OFF.

Big idea
Electricity only travels when the circle is not broken.

Our builder's kit

Meet the parts



LED

Our tiny light. It has two legs.



Coin cell

A tiny round battery. The energy box.



Copper tape

Shiny sticky road for electricity.



Card + markers

Your design. Make it yours!

Conductor: a material that lets electricity through. Metal like copper is a great conductor. Paper is not!

LED means

Light Emitting Diode. A grown-up word for a tiny light!

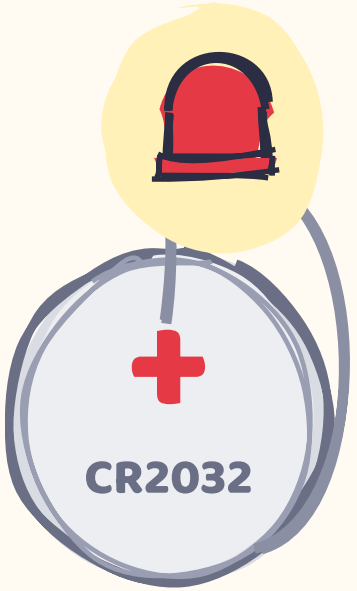
MAKE

Be a scientist!

Guess, then test

Hold the LED on the coin cell

Pinch the battery between the two legs. Long leg on the **+ side**.
Then **flip** the LED round. What happens now?



Test	My guess	What happened?
Long leg on +	ON / OFF	ON / OFF
LED flipped round	ON / OFF	ON / OFF
Only one leg touching	ON / OFF	ON / OFF

Surprise!
An LED is a one-way street. Electricity can only go in one direction!

Time to build!

Make your card in 6 steps

1 **Draw** your picture on the card. Mark a dot where the light goes.

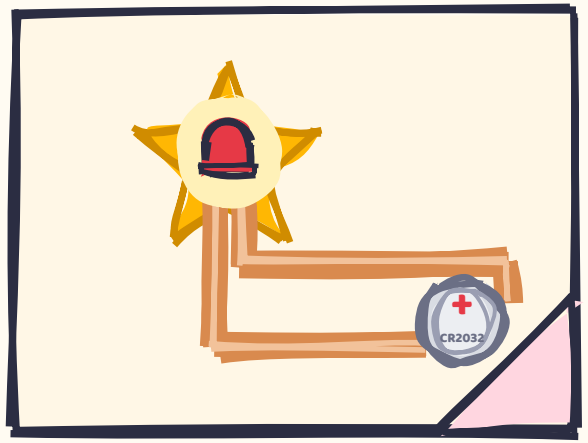
2 **Stick** two copper tape roads from the dot to the corner. Do not let them touch!

3 **Place** the LED legs flat on the tape. Long leg on the + road.

4 **Sit** the coin cell on the - road, + side up.

5 **Fold** the corner flap so the + road touches the top of the battery.

6 **Press** the corner and watch your card glow!



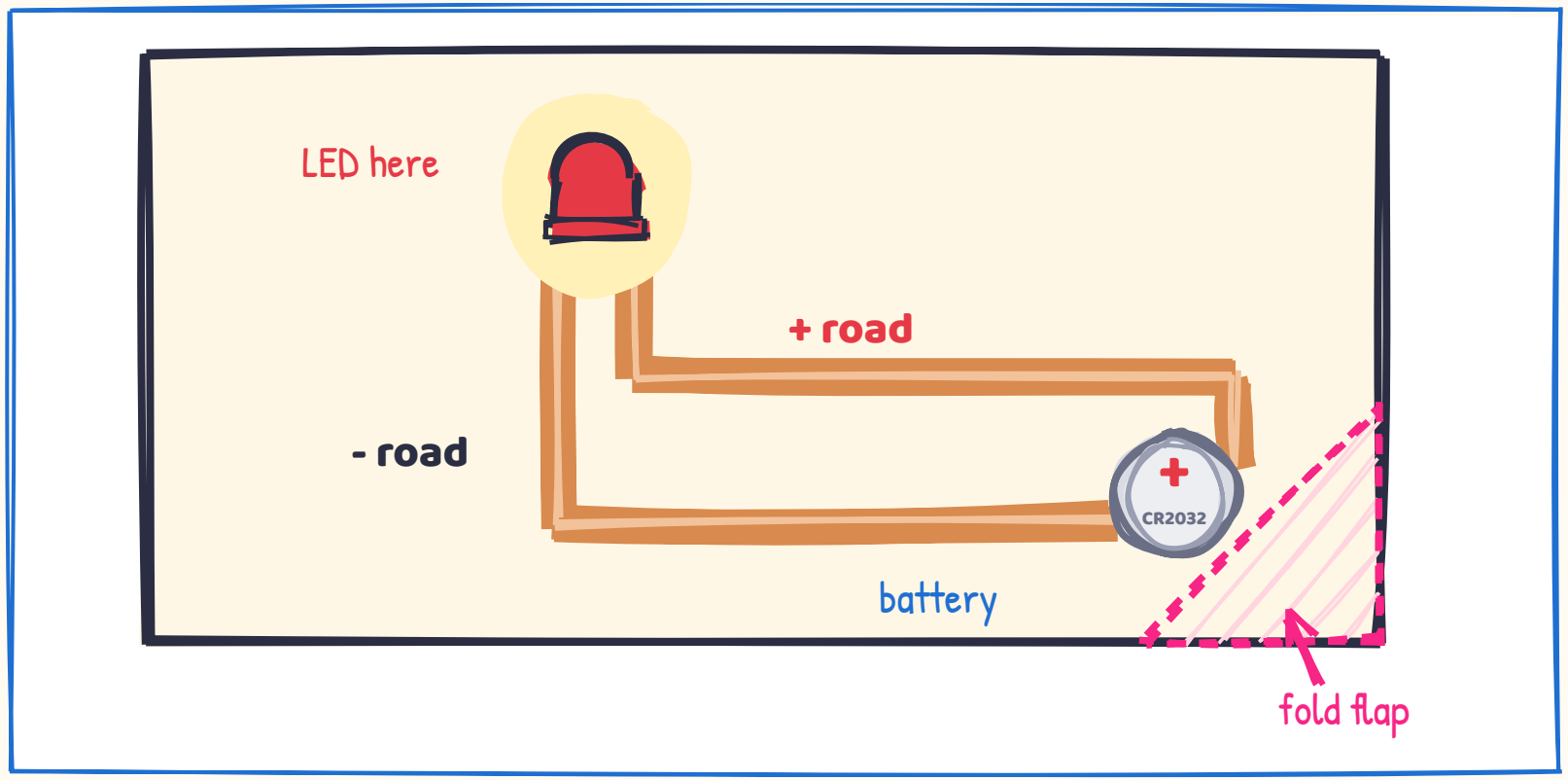
your finished card

Corner tip
Do not cut the tape at a corner. Fold it like a ribbon!

MAKE

Look inside

How your card is wired



Follow the road

+ side of battery
along the + road
into the LED long leg
out of the short leg
along the - road
back home to the battery!

Teacher: print the template on card. One coin cell lights one or two LEDs; for more, use a fresh cell.

No light? Don't worry!

Be a circuit detective

1. Is there a **gap** in the tape? Add a small tape bridge and press it flat.
2. Is the LED **the wrong way**? Flip it! Long leg on the + road.
3. Are the LED legs **touching** the tape? Tape them down firmly.
4. Do the two roads **touch each other**? That is a short cut! Pull them apart.



Real engineers

Every engineer finds bugs. Fixing them is the fun part!

Finished early?

Explorer challenges

Challenge time!



Open-to-glow: move the battery into the fold so the light turns on when the card opens.



Double shine: add a second LED on the same roads. Do both glow?



Light it up: design a firefly, a rocket, a lighthouse or a star that glows.

Show and tell

Hold up your card and finish the sentence:

"My light is ON because the loop is …"

A gift!

Who will you give your card to? Mum, Dad, a grandparent, a teacher?

Show what you know

What did we learn?

- [x] A circuit is a closed loop, like a circle road
- [x] A gap breaks the loop and the light goes OFF
- [x] LED: long leg is plus (+)
- [x] Copper is a conductor, paper is not



Circuit Explorer
badge unlocked!

At home
Find 3 things that use a battery. Tell us next class!

Answer the mystery: it was not magic. It was a **closed circuit**!

Next class: a Push-Button Doorbell. Ding dong!